Reverse-check review package — `MDMmodel_gen`

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	`MDMmodel/repair/final.fr`
original model name	`MDMmodel_gen` (hidden from the agent)
paper	MDMmodel/text/1311.6661.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

`LFermionsDM` (`:=`)

```
Block[{mu, sp, cc}, ExpandIndices[I TLbar.Ga[mu].DC[TL, mu] + I TLbar.Ga[mu].DC[TR, mu] + I
↪  TRbar.Ga[mu].DC[TL, mu] + I TRbar.Ga[mu].DC[TR, mu] - Mdltn/vevf S (TLbar[sp, cc].TL[sp,
↪  cc] + TLbar[sp, cc].TR[sp, cc] + TRbar[sp, cc].TL[sp, cc] + TRbar[sp, cc].TR[sp, cc]),
↪  FlavorExpand -> {SU2W, SU2D}]/.{CKM[a_, b_] Conjugate[CKM[a_, c_]] -> IndexDelta[b, c],
↪  CKM[b_, a_] Conjugate[CKM[c_, a_]] -> IndexDelta[b, c]}]
```

`LHiggsMDM` (`:=`)

```
Block[{ii, jj, mu, feynmangaugerules}, feynmangaugerules = If[Not[FeynmanGauge], {G0 | GP |
↪  GPbar -> 0}, {}]; ExpandIndices[DC[Phibar[ii], mu] DC[Phi[ii], mu] - 1/2 del[S, mu]
↪  del[S, mu] - muS2/2 S^2 - dlams/24 S^4 - dkappa/2 S^2 Phibar[ii] Phi[ii] - muH2
↪  Phibar[ii] Phi[ii] - dlamh/4 Phibar[ii] Phi[ii] Phibar[jj] Phi[jj], FlavorExpand ->
↪  {SU2D, SU2W}]/.feynmangaugerules]
```

`LYukawaDM` (`:=`)

```
Block[{sp, ii, jj, cc, yuk, feynmangaugerules}, feynmangaugerules = If[Not[FeynmanGauge], {G0
↪  | GP | GPbar -> 0}, {}]; yuk = -yp QLb[sp, ii, 3, cc].TR[sp, cc] Phibar[jj] Eps[ii, jj];
↪  yuk = ExpandIndices[yuk + HC[yuk], FlavorExpand -> SU2D]; yuk = yuk /. {CKM[a_, b_]
↪  Conjugate[CKM[a_, c_]] -> IndexDelta[b, c], CKM[b_, a_] Conjugate[CKM[c_, a_]] ->
↪  IndexDelta[b, c]}; yuk/.feynmangaugerules]
```

`LMDMNP` (`:=`)

```
LFermionsDM + LHiggsMDM + LYukawaDM
```

`LTot` (`:=`)

```
LFermionsDM + LHiggsMDM + LYukawaDM
```

Blank-slate reconstruction

Reconstructed Physics From `sanitized.fr`

Lagrangian

Field definitions used by the Lagrangian:

$$S = v_f + s_s\,h + c_s\,s_{\mathrm{DM}}$$

$$\Phi = \begin{pmatrix} -iG^+ \\ \dfrac{v + c_s h - s_s s_{\mathrm{DM}} + iG^0}{\sqrt{2}} \end{pmatrix}$$

$$T_L = P_L(-s_l\,t + c_l\,t'), \qquad T_R = P_R(-s_r\,t + c_r\,t')$$

with

$$P_L = \frac{1-\gamma^5}{2}, \qquad P_R = \frac{1+\gamma^5}{2}.$$

The covariant derivatives are

$$D_\mu\Phi = \left(\partial_\mu - ig\,W_\mu^a\frac{\sigma^a}{2} - ig'\frac{1}{2}B_\mu\right)\Phi,$$

$$D_\mu T = \left(\partial_\mu - ig_s\,G_\mu^A T^A - ig'\frac{2}{3}B_\mu\right)T,$$

where T is an $SU(3)_c$ triplet, $SU(2)_L$ singlet fermion with $Y = Q = 2/3$. The singlet scalar S has ordinary derivative $\partial_\mu S$.

`LFermionsDM`

$$\mathcal{L}^{\texttt{LFermionsDM}}_{\mathrm{kin},T} = i\,\overline{T_L}\gamma^\mu D_\mu T_L + i\,\overline{T_L}\gamma^\mu D_\mu T_R + i\,\overline{T_R}\gamma^\mu D_\mu T_L + i\,\overline{T_R}\gamma^\mu D_\mu T_R.$$

Equivalently, with $T = T_L + T_R$,

$$\mathcal{L}^{\texttt{LFermionsDM}}_{\mathrm{kin},T} = i\,\overline{T}\gamma^\mu D_\mu T.$$

$$\mathcal{L}^{\texttt{LFermionsDM}}_{S\overline{T}T} = -\frac{M_\Delta}{v_f}\,S\,\left(\overline{T_L}T_L + \overline{T_L}T_R + \overline{T_R}T_L + \overline{T_R}T_R\right),$$

with

$$M_\Delta \equiv \texttt{Mdltn} = M_{T'}c_l.$$

Equivalently,

$$\mathcal{L}^{\texttt{LFermionsDM}}_{S\overline{T}T} = -\frac{M_\Delta}{v_f}\,S\,\overline{T}T.$$

`LHiggsMDM`

$$\mathcal{L}^{\texttt{LHiggsMDM}}_{\Phi,\mathrm{kin}} = (D_\mu\Phi)^\dagger(D^\mu\Phi).$$

$$\mathcal{L}^{\texttt{LHiggsMDM}}_{S,\mathrm{kin}} = -\frac{1}{2}\,\partial_\mu S\,\partial^\mu S.$$

$$\mathcal{L}^{\texttt{LHiggsMDM}}_{S^2} = -\frac{\mu_S^2}{2}\,S^2.$$

$$\mathcal{L}^{\texttt{LHiggsMDM}}_{S^4} = -\frac{\lambda_S}{24}\,S^4.$$

$$\mathcal{L}^{\texttt{LHiggsMDM}}_{S^2\Phi^\dagger\Phi} = -\frac{\kappa}{2}\,S^2\,\Phi^\dagger\Phi.$$

$$\mathcal{L}^{\texttt{LHiggsMDM}}_{\Phi^\dagger\Phi} = -\mu_H^2\,\Phi^\dagger\Phi.$$

$$\mathcal{L}^{\texttt{LHiggsMDM}}_{(\Phi^\dagger\Phi)^2} = -\frac{\lambda_H}{4}\,(\Phi^\dagger\Phi)^2.$$

The parameters appearing here are the internal FeynRules symbols

$$\kappa = \texttt{dkappa}, \qquad \lambda_H = \texttt{dlamh}, \qquad \lambda_S = \texttt{dlams},$$

$$\mu_S^2 = \texttt{muS2}, \qquad \mu_H^2 = \texttt{muH2}.$$

`LYukawaDM`

The FeynRules expression

$$-\texttt{yp}\,\overline{Q}^i_{L,3}\,T_R\,\Phi^{\dagger j}\,\epsilon_{ij} + \text{h.c.}$$

is

$$\mathcal{L}^{\texttt{LYukawaDM}}_{\mathrm{Yuk}} = -y'\,\overline{Q}^i_{L,3}\,\widetilde{\Phi}_i\,T_R + \text{h.c.},$$

where

$$\widetilde{\Phi}_i \equiv \epsilon_{ij}\Phi^{\dagger j}, \qquad Q_{L,3} = \begin{pmatrix} t_L \\ b_L \end{pmatrix},$$

and

$$y' = \texttt{yp} = \frac{\sqrt{2}}{v}M_{T'}s_l.$$

The hermitian conjugate is

$$-y'^*\,\overline{T_R}\,\widetilde{\Phi}^\dagger_i Q^i_{L,3}.$$

Field Table

.fr class	Symbol	Spin	SU(3) rep	SU(2) rep	$U(1)_Y$ / charge	Self-conjugate	Mass
`S[21]`	`sDM`	0	singlet	singlet	$Y = 0,\ Q = 0$	yes	$M_{s\mathrm{DM}} =$ `MsDM` $=$ 173.2
`F[7]`	`tp`	1/2	triplet	singlet	$Y = 2/3,\ Q = 2/3$	no	$M_{T'} =$ `MTP` $=$ 1670.3
`S[31]`	`S`	0	singlet	singlet	$Y = 0,\ Q = 0$	yes	unphysical gauge/electroweak-basis field
`S[11]`	`Phi`	0	singlet	doublet	$Y = 1/2$	no	unphysical Higgs doublet
`F[26]`	`TR`	1/2	triplet	singlet	$Y = 2/3,\ Q = 2/3$	no	unphysical right-chiral component
`F[27]`	`TL`	1/2	triplet	singlet	$Y = 2/3,\ Q = 2/3$	no	unphysical left-chiral component

External Parameters

Symbol	Value	Appears in	Physical meaning
`eta`	0.33	$v_f = v/\eta$	Ratio setting the singlet vev v_f relative to the SM Higgs vev v.
`ts`	-0.23	$s_s = t_s/\sqrt{1+t_s^2}$, $c_s = 1/\sqrt{1+t_s^2}$, scalar potential parameters	Tangent of the scalar mixing angle, $t_s = \tan\theta_S$.
`sl`	0.12	$c_l = \sqrt{1-s_l^2}$, $s_r = M_t s_l/(M_{T'} c_l)$, $y' = \sqrt{2} M_{T'} s_l/v$, $M_\Delta = M_{T'} c_l$	Sine of the left-handed top-partner mixing angle, $s_l = \sin\theta_L$.

Physics Summary

The file encodes a singlet-scalar extension of the Higgs sector together with a vector-like color-triplet, electroweak-singlet fermion of charge 2/3 that mixes chirally with the SM top quark. The physical scalar `sDM` is a real scalar mixed with the SM Higgs through the singlet vev and portal interaction, while `tp` is a heavy top partner coupled to the third-generation quark doublet through a $\widetilde{\Phi}$ Yukawa interaction. The model mediates Higgs-portal scalar interactions, scalar production and decay through mixing, and top-partner processes such as t' production and decays involving t, h, electroweak bosons, and the singlet-like scalar.

Paper cross-check

Located Paper Definitions

The paper's model definition is in Section II, "The Minimal Dilaton Model." It defines the field content as one gauge-singlet scalar S and one fermion T with the same quantum numbers as the right-handed top quark. The effective Lagrangian is Eq. (2), the scalar potential is Eq. (3), scalar mixing is Eq. (4), the scalar-potential reparameterization is Eq. (5), η is Eq. (8), and the left-handed top/top-partner mixing is Eq. (9). Eq. (1) is background for traditional dilaton couplings and is explicitly not the distinctive MDM setup used later.

paper term (eq. ref)	reconstruction term	verdict	notes
Field content: gauge-singlet scalar S, fermion T with right-handed-top quantum numbers, Section II before Eq. (2)	S singlet scalar; T color triplet, $SU(2)_L$ singlet, $Y = Q = 2/3$	agree	"Same quantum number as right-handed top quark" implies the listed T gauge representation.
L_{SM} without Higgs potential, Eq. (2)	Not reconstructed as full SM; only Higgs kinetic/covariant derivative and third-generation/top-partner pieces are shown	missing-in-reconstruction	The paper includes all SM kinetic, gauge, and Yukawa terms except the Higgs potential. The reconstruction is only the BSM/model-relevant subset.
Singlet kinetic term $-\frac{1}{2}\partial_\mu S \partial^\mu S$, Eq. (2)	$-\frac{1}{2}\partial_\mu S \partial^\mu S$	agree	Same normalization and sign, modulo metric convention.
Top-partner kinetic term, schematically $-\bar{T} slashed D T$, Eq. (2)	$i\bar{T}\gamma^\mu D_\mu T$, expanded through T_L, T_R	agree	Same physics up to common extraction/sign conventions for the Dirac kinetic term. The reconstruction's cross-chiral expansion is redundant; the compact $i\bar{T}\gamma^\mu D_\mu T$ is the meaningful form.
Dilaton/top-partner mass interaction $-\bar{T}(M/f)ST$, Eq. (2)	$-M_\Delta/v_f\, S\bar{T}T$, with $M_\Delta = M_{T'} c_l$	agree	Same structure if $v_f = f$ and the implementation identifies the paper's M with M_Δ. That mass-parameter mapping is not stated explicitly in the paper excerpt.
Yukawa interaction $-[y'\,\bar{T}_R(q_{3L} \cdot H) + \text{h.c.}]$, Eq. (2)	$-y' \bar{Q}^i_{L,3} \widetilde{\Phi}_i T_R + \text{h.c.}$	agree	Equivalent physics when written in the hermitian-conjugate orientation with the usual ϵ_{ij} contraction/$\widetilde{\Phi}$ convention.
q_{3L} is the third-generation $SU(2)_L$ left-handed quark doublet, text after Eq. (2)	$Q_{L,3} = (t_L, b_L)^T$	agree	Same field content.

paper term (eq. ref)	reconstruction term	verdict	notes
Scalar potential $+\frac{m_S^2}{2}S^2+\frac{\lambda_S}{4!}S^4+\frac{\kappa}{2}S^2\lvert H\rvert^2+m_H^2\lvert H\rvert^2+\frac{\lambda_H}{4}\lvert H\rvert^4$ inside $\widetilde{V}$, Eq. (3), entering the Lagrangian as $-\widetilde{V}$	$-\frac{\mu_S^2}{2}S^2-\frac{\lambda_S}{24}S^4-\frac{\kappa}{2}S^2\Phi^\dagger\Phi-\mu_H^2\Phi^\dagger\Phi-\frac{\lambda_H}{4}(\Phi^\dagger\Phi)^2$	agree	Coefficients match once the reconstruction's $\mu_{S,H}^2$ are identified with the paper's $m_{S,H}^2$, and $\lvert H\rvert^2=\Phi^\dagger\Phi$.
Higgs doublet neutral component $H^0=\frac{1}{\sqrt{2}}(v+h\cos\theta_S-s\sin\theta_S)$, Eq. (4)	$\Phi^0=\frac{1}{\sqrt{2}}(v+c_sh-s_ss_{\mathrm{DM}}+iG^0)$	agree	Same neutral scalar mixing if $c_s=\cos\theta_S$, $s_s=\sin\theta_S$, and $s_{\mathrm{DM}}=s$. Goldstone terms are gauge-completion details absent from the paper equation.
Singlet mixing $S=f+h\sin\theta_S+s\cos\theta_S$, Eq. (4)	$S=v_f+s_sh+c_ss_{\mathrm{DM}}$	agree	Same mixing if $v_f=f$, $s_s=\sin\theta_S$, $c_s=\cos\theta_S$.
Scalar-potential parameter relations for $\kappa,\lambda_H,\lambda_S$, Eq. (5)	Reconstruction lists internal $\kappa,\lambda_H,\lambda_S,\mu_S^2,\mu_H^2$, but does not reproduce the Eq. (5) formulas	missing-in-reconstruction	The Lagrangian terms are present, but the paper's coefficient relations in terms of f,v,θ_S,m_h,m_s, including absolute values and $\mathrm{Sign}(\sin 2\theta_S)$, are not fully reconstructed.
$\eta\equiv v/(fN_T)$, with $N_T=1$ for MDM, Eq. (8)	$v_f=v/\eta$	agree	Equivalent for $N_T=1$.
Left-handed mixing $q_{3L}^u=\cos\theta_L\,t_L+\sin\theta_L\,t'_L$, Eq. (9)	Not given; reconstruction only defines $Q_{L,3}=(t_L,b_L)^T$	missing-in-reconstruction	The paper explicitly gives the gauge-basis upper component of q_{3L} in terms of mass eigenstates.
Left-handed top-partner mixing $T_L=-\sin\theta_L\,t_L+\cos\theta_L\,t'_L$, Eq. (9)	$T_L=P_L(-s_lt+c_lt')$	agree	Same relation if $s_l=\sin\theta_L$, $c_l=\cos\theta_L$.
Right-handed top-partner mixing	$T_R=P_R(-s_rt+c_rt')$	extra-in-reconstruction	Section II of the paper does not explicitly define a right-handed mixing angle. This may be an implementation detail inferred from mass diagonalization or from Ref. [15], but it is not in the displayed paper definition.
Full Higgs doublet with charged/neutral Goldstones	$\Phi=(-iG^+,\,(v+c_sh-s_ss_{\mathrm{DM}}+iG^0)/\sqrt{2})^T$	extra-in-reconstruction	The paper only displays H^0 in Eq. (4). The Goldstone and phase conventions are standard gauge-basis completion, not a contradiction.
Covariant derivative of Φ with $Y=1/2$	Explicit $D_\mu\Phi=(\partial_\mu-igW_\mu^a\sigma^a/2-ig'B_\mu/2)\Phi$	agree	Consistent with the SM Higgs doublet contained in L_{SM}.

paper term (eq. ref)	reconstruction term	verdict	notes
Covariant derivative of T with color triplet, $SU(2)$ singlet, $Y = 2/3$	Explicit $D_\mu T = (\partial_\mu - ig_s G^A T^A - ig'(2/3)B_\mu)T$	agree	Consistent with “same quantum number as the right-handed top quark.”

Disagreements And Checks

- **Missing full $L_{\rm SM}$ content**: severity **substantive**. A human should check whether the implementation intentionally imports the SM Lagrangian elsewhere, since the reconstruction itself only shows the BSM/Higgs-sector subset.
- **Missing Eq. (5) scalar-parameter relations**: severity **substantive**. A human should check the implementation definitions of $\mu_S^2, \mu_H^2, \lambda_H, \lambda_S, \kappa$ against the paper’s formulas, especially the sign and absolute-value conventions.
- **Missing q_{3L}^u left-handed mixing relation from Eq. (9)**: severity **substantive**. A human should check whether the implementation rotates the SM top component consistently with $q_{3L}^u = c_L t_L + s_L t'_L$, not only the heavy T_L component.
- **Extra right-handed mixing $T_R = P_R(-s_r t + c_r t')$**: severity **convention**. A human should check whether this is an implementation-level diagonalization convention compatible with the paper’s approximations $m_{t'} \gg m_t$ and $\tan\theta_L \ll m_{t'}/m_t$.
- **Extra Goldstone/full-doublet conventions**: severity **cosmetic**. A human should check only that the $-iG^+$ and $+iG^0$ phase conventions are used consistently in the implementation’s gauge fixing and Feynman rules.
- **Top-partner mass mapping $M \leftrightarrow M_\Delta = M_{T'} c_l$**: severity **convention**. A human should verify the implementation’s input-parameter definitions, because the paper names M as the strong-dynamics scale but does not explicitly state the reconstructed $M_{T'} c_l$ relation in Eq. (2).

Overall Assessment

The reconstruction captures the central MDM Lagrangian structure from Section II: a real singlet scalar mixed with the neutral Higgs, the scalar potential with the same operator content and normalizations, a vector-like $SU(3)_c$ triplet $SU(2)_L$ singlet top partner of charge 2/3, its dilaton-proportional mass interaction, and the y' Yukawa mixing with the third-generation quark doublet. The main gaps are not obvious wrong operators but incomplete reconstruction of paper-level definitions: the full $L_{\rm SM}$ context, the Eq. (5) coefficient relations, and the full left-handed top mixing relation for q_{3L}^u. The extra right-handed rotation and Goldstone-expanded Higgs doublet look like implementation conventions rather than direct conflicts, but they should be checked against the implementation’s mass diagonalization and gauge conventions.

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field’s representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 18 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model’s identity.

Physicist sign-off

- Reviewed by: ________________________________ Date: __________________

- Verdict: [ ] approve [ ] approve with corrections [ ] reject
- Notes: